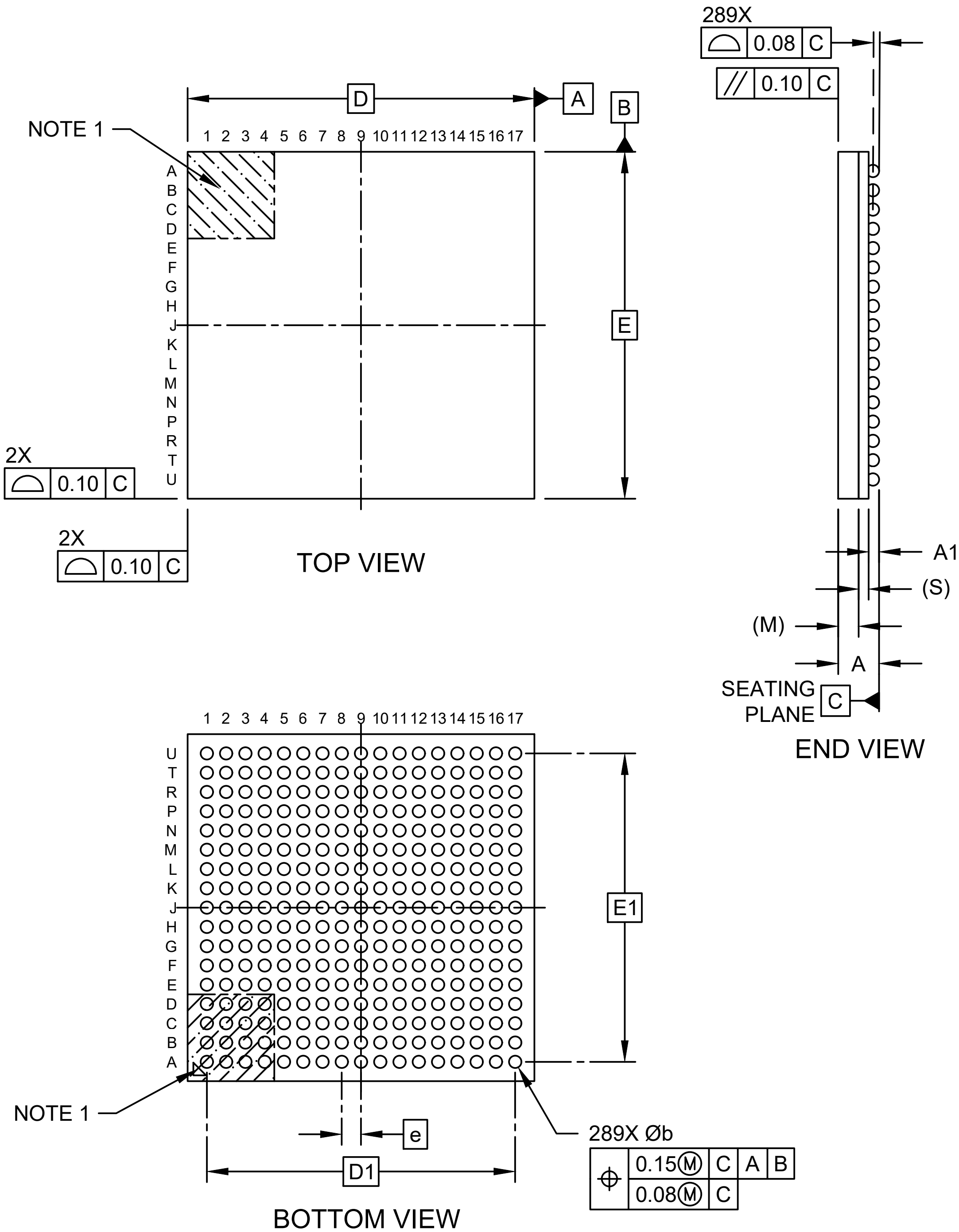
289-Ball Thin Fine-Pitch Ball Grid Array Package (8GW) - 9x9x1.2 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



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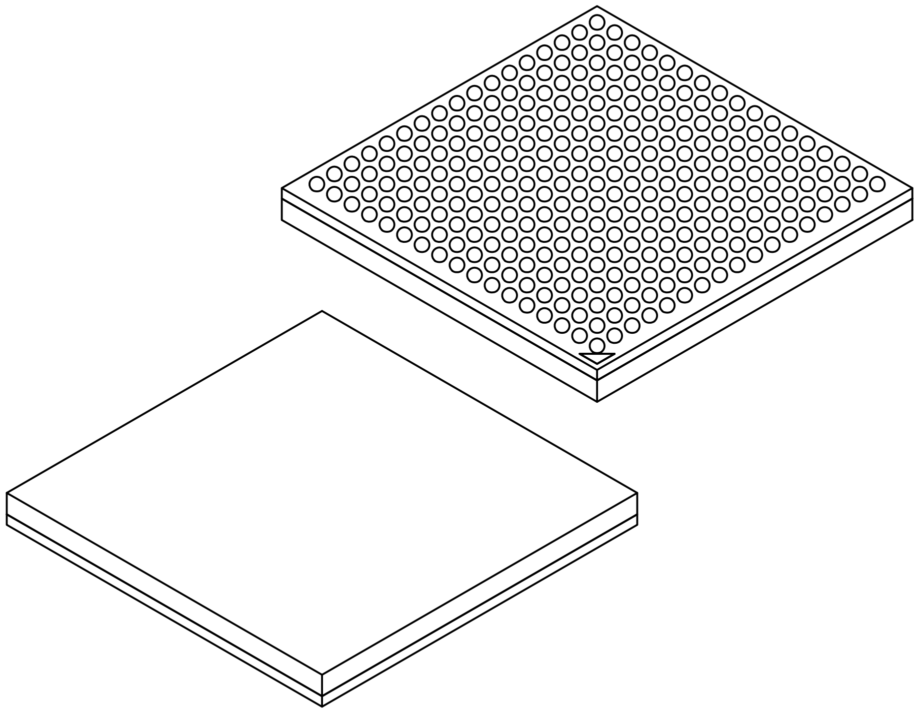
Packaging Diagrams and Parameters

289-Ball Thin Fine-Pitch Ball Grid Array Package (8GW) - 9x9x1.2 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		289	
Pitch	e		0.50 BSC	
Overall Height	A	–	–	1.200
Ball Height	A1	0.160	0.210	0.260
Mold Thickness	M		0.530 REF	
Substrate Thickness	S		0.352 REF	
Overall Length	D		9.00 BSC	
Ball Array Length	D2		8.00 BSC	
Overall Width	E		9.00 BSC	
Ball Array Width	E2		8.00 BSC	
Ball Diameter	b	0.270	0.320	0.370

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

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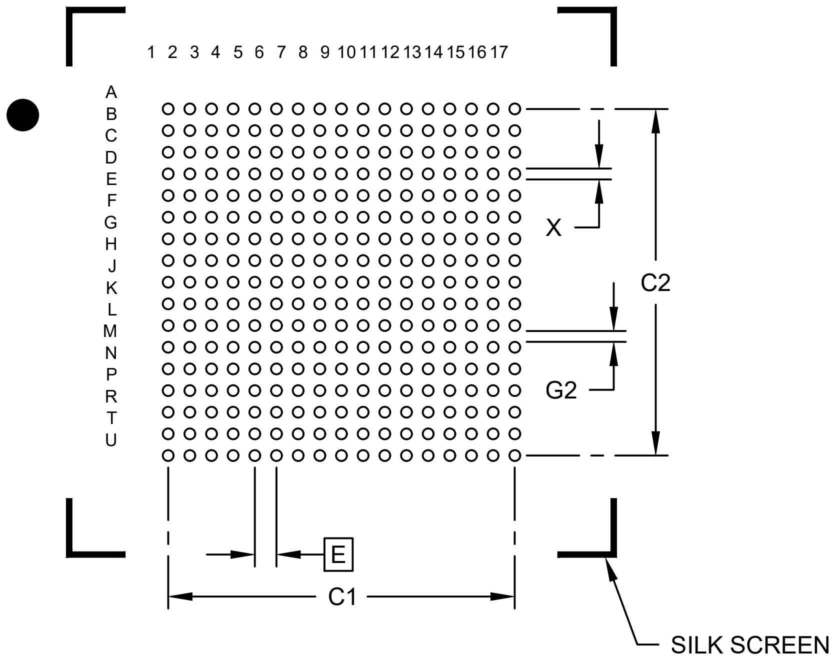
289-Ball Thin Fine-Pitch Ball Grid Array Package (8GW) - 9x9x1.2 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		8.00	
Contact Pad Spacing	C2		8.00	
Contact Pad Diameter (X289)	X			0.25
Contact Pad to Contact Pad (X285)	G2	X.XX		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, pleease refer to current industry standard IPC-7093.

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